

Investigation of Gold and Silver Nanoparticles on Absorption Heating and Scattering Imaging

Yu Juan Zhang

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Abstract Efficient conversion of absorbed light to heat energy and strong scattering by gold and silver nanoparticles suggest these nanoparticles as the agents of heating and imaging. Absorption efficiency and scattering efficiency of gold and silver nanoparticles were studied through numerical simulation using the discrete dipole approximation method. This study shows that the size of gold and silver nanoparticles can effect gold and silver nanoparticles' absorption efficiency and scattering efficiency. The gold nanoparticle is found to possess the maximum absorption efficiency when the size of gold nanoparticle is 50 nm and the incident wavelength is 540 nm, and the increasing scattering efficiency with the increasing size of gold nanoparticle in the medium, and refractive index of the medium is around 1.33. However, the silver nanoparticle owns the maximum absorption efficiency when the size of silver nanoparticle is 20 nm and the incident wavelength is 396 nm, and the maximum scattering efficiency when the size of silver nanoparticle is 30 nm and the incident wavelength is 410 nm in the same medium. The conditions for achieving the maximum adsorption efficiency and scattering efficiency of gold and silver nanoparticle can be used for heating and imaging using visible and near-infrared light.

Keywords Nanoparticles · Absorption efficiency · Scattering efficiency · Heating · Imaging

Introduction

Gold and silver nanoparticles have attracted considerable attention because of their special physical and chemical properties compared to the bulk counterparts. One of the most interesting aspects is their optical properties. On the nanoscale, gold and silver exhibit strong absorption and scattering in the visible and near-infrared region of the spectrum because of surface plasmon resonance (SPR), and the origin of SPR is attributed to coherent interaction of the electrons in the metal in the conduction band with the electromagnetic field of light which results in the electrons in the metal to oscillate [1]. The optical properties of nanoparticles can be changed by the shape or material of nanoparticle and the refractive index around the nanoparticle [2–4]. The controllable and tunable optical properties of gold and silver nanoparticles were highly desirable for many applications, including imaging [2, 5], photodynamic therapy [5–7] and sensing [1].

Light absorption appears when the photon energy is dissipated on account of the electron oscillation in the nanoparticle induced by the photon, which has been efficiently converted to heat energy. Nanoparticles have been used as selective photothermal agent in molecular cancer cell targeting by heating, which were affected by the size of nanoparticles [6, 7]. Light scattering occurs when the photon causes the electron oscillation in the nanoparticle which emit photons in the form of scattered light either at the same frequency as the incident light (Rayleigh scattering) or at a shifted frequency (Raman scattering). Preliminary studies have reported that gold nanoparticles used as contrast agents for biomedical imaging using confocal–multiphoton plasmon resonance microscopy [8], scanning optical microscopy [9] and third-harmonic microscopy [10]. Different images were displayed because of

Y. J. Zhang (✉)
Department of Pharmaceutics, Biopharmaceutics
and Biotechnology, Free University of Berlin,
Kelchstr. 31,
12169 Berlin, Germany
e-mail: zhangyujuan@zedat.fu-berlin.de

changes in the structure and size of silver nanoparticles [2]. However, it is still necessary to explore the effects of the size of nanoparticles on the thermal therapy and imaging.

In this paper, the discrete dipole approximation (DDA) method is used to calculate the absorption efficiency and scattering efficiency of gold and silver nanoparticle in the medium. The correlation of the absorption efficiency and scattering efficiency of nanoparticles with the size of nanoparticles is discussed. This paper tries to analyse the relationship between absorption efficiency of nanoparticle and heating, as well as scattering efficiency of nanoparticle and imaging using DDA method. The final aim of studies is to better apply nanoparticles in therapy and imaging.

DDA Simulation Section

Numerical simulation is performed using the DDA algorithm, which is a powerful calculation method for calculating absorption efficiency, scattering efficiency and extinction efficiency for the targets of arbitrary structures [11, 12]. The target is represented as a lattice of polarisable cubic elements (N -point dipoles) whose positions and polarisabilities are signed as r_i and α_i , $i=1,2,\dots,N$. And the target is excited by a monochromatic incident plane wave. The polarisation induced in each dipole can be expressed as:

$$P_i = \alpha_i \cdot E_{loc,i}(r_i) \quad (1)$$

where the local field $E_{loc,i}(r_i)$ is the sum of the incident radiation field and the field radiated by the other $N-1$ dipoles. For a given incident wavelength λ , the field can be expressed as:

$$E_{loc,i}(r_i) = E_{inc,i} + E_{dip,i} = E_0 \exp(ik \cdot r_i) - \sum_{j \neq i} A_{ij} \cdot P_j \quad (2)$$

where E_0 and $k = \frac{2\pi}{\lambda}$ are the amplitude and wave number of the incident wave, respectively. Then, the interaction matrix A can be expressed as:

$$A_{ij} \cdot P_j = \frac{\exp(ik|r_{ij}|)}{|r_{ij}|^3} \left\{ k^2 r_{ij} \times (r_{ij} \times P_j) + \frac{(1 - ik|r_{ij}|)}{|r_{ij}|^2} \times [|r_{ij}|^2 P_i - 3r_{ij}(r_{ij} \cdot P_j)] \right\} \quad (3)$$

($j = 1, 2, \dots, N, j \neq i$)

where $r_{ij} = r_i - r_j$ and $|r_{ij}| = |r_i - r_j|$. Substituting Eq. 2 and Eq. 3 into Eq. 1 and rearranging Eq. 1 generate an

equation of the form:

$$A' \cdot P = E \quad (4)$$

where A' is a matrix which is built out of the matrix A from Eq. 3. For a target with a total of N dipoles, E and P in Eq. 4 are $3N$ -dimensional vectors, and A' is a $3N \times 3N$ matrix. By solving these $3N$ linear equations, the polarisation vector P can be obtained. Then, the extinction and absorption cross sections for the target can be calculated by:

$$C_{ext} = \frac{4\pi k}{|E_0|^2} \sum_{i=1}^N \text{Im}(E_{loc,i}^* \cdot P_i) \quad (5)$$

$$C_{abs} = \frac{4\pi k}{|E_0|^2} \sum_{i=1}^N \left\{ \text{Im}[P_i \cdot (\alpha_i^{-1})^* P_i^*] - \frac{2k^3}{3} |P_i|^2 \right\} \quad (6)$$

where $*$ means complex conjugate. The scattering cross section can be obtained using the following relation:

$$C_{ext} = C_{sca} + C_{abs} \quad (7)$$

For calculations, a free code named DDSCAT 7.1.0 was used, which was developed by Draine and Flatau [13]. Depending on the error tolerance in the calculation for gold and silver nanoparticles, the typical cube size of each dipole is lesser than 1.5 nm, and the number of dipoles (N) is between 1.4×10^4 and 3.4×10^4 . The size of nanoparticles is in the range of 10–60 nm, and the structure of nanoparticles is spherical in the calculation. The dielectric constants for gold and silver are taken from reference [14]. In the calculation, gold and silver nanoparticles are in the medium, and the refractive index of the medium is around 1.33. The incident wavelength is varied from 300 to 850 nm.

The extinction efficiency (Q_{ext}), absorption efficiency (Q_{abs}) and scattering efficiency (Q_{sca}) are calculated by DDSCAT 7.1.0. Q_{ext} , Q_{abs} and Q_{sca} are expressed as:

$$Q_{ext} = \frac{C_{ext}}{\pi a_{eff}^2}, \quad Q_{abs} = \frac{C_{abs}}{\pi a_{eff}^2}, \quad Q_{sca} = \frac{C_{sca}}{\pi a_{eff}^2} \quad (8)$$

where a_{eff} is defined through the concept of an effective volume equal to $\frac{4\pi a_{eff}^3}{3}$ in the calculation.

Results and Discussions

In order to analyse the relationship between absorption efficiency of nanoparticle and heating, as well as scattering efficiency of nanoparticle and imaging, optical efficiencies for sphere of gold and silver nanoparticles were calculated using

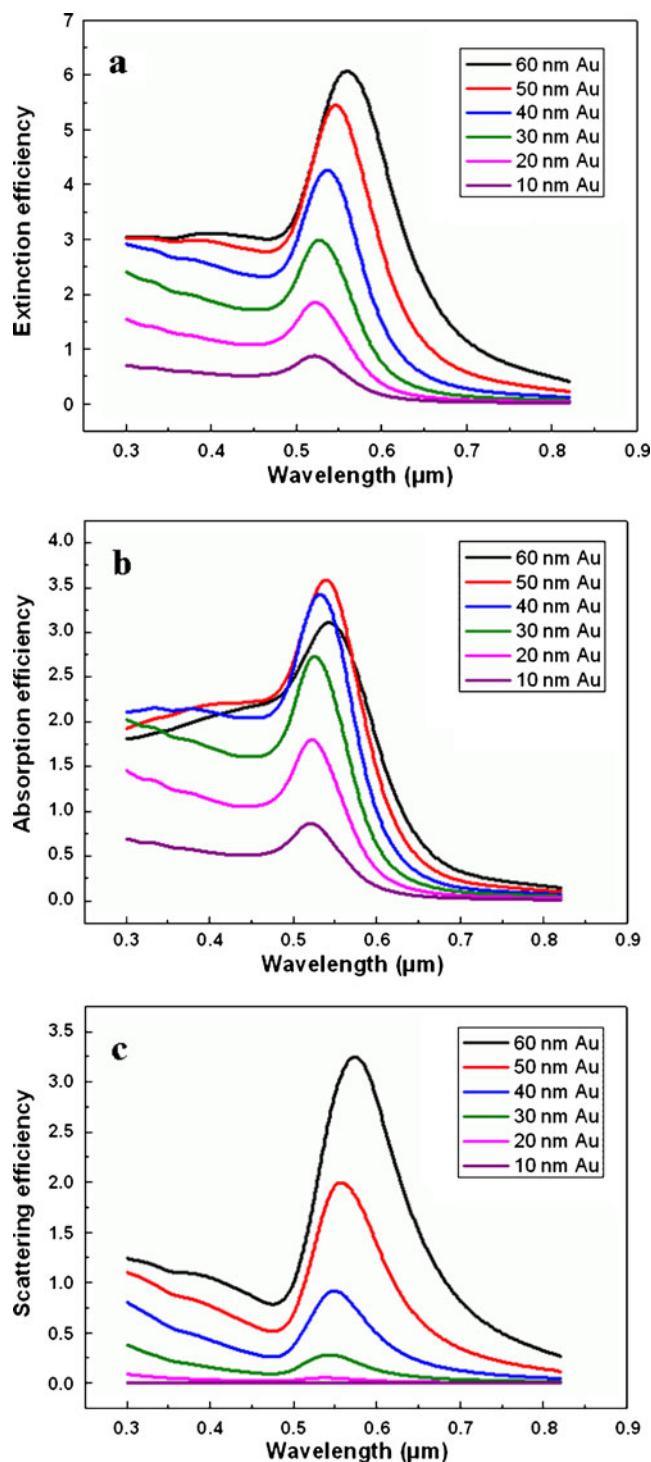


Fig. 1 DDA simulation of optical properties of spherical gold nanoparticles surrounded with the medium, and the refractive index of the medium is 1.33. **a** Extinction efficiencies. **b** Absorption efficiencies. **c** Scattering efficiencies

DDA method, and the size of nanoparticle is in the range of 10–60 nm. The absorption heating efficiency and scattering imaging efficiency are defined as absorption efficiency (Q_{abs}) and scattering efficiency (Q_{sca}), respectively.

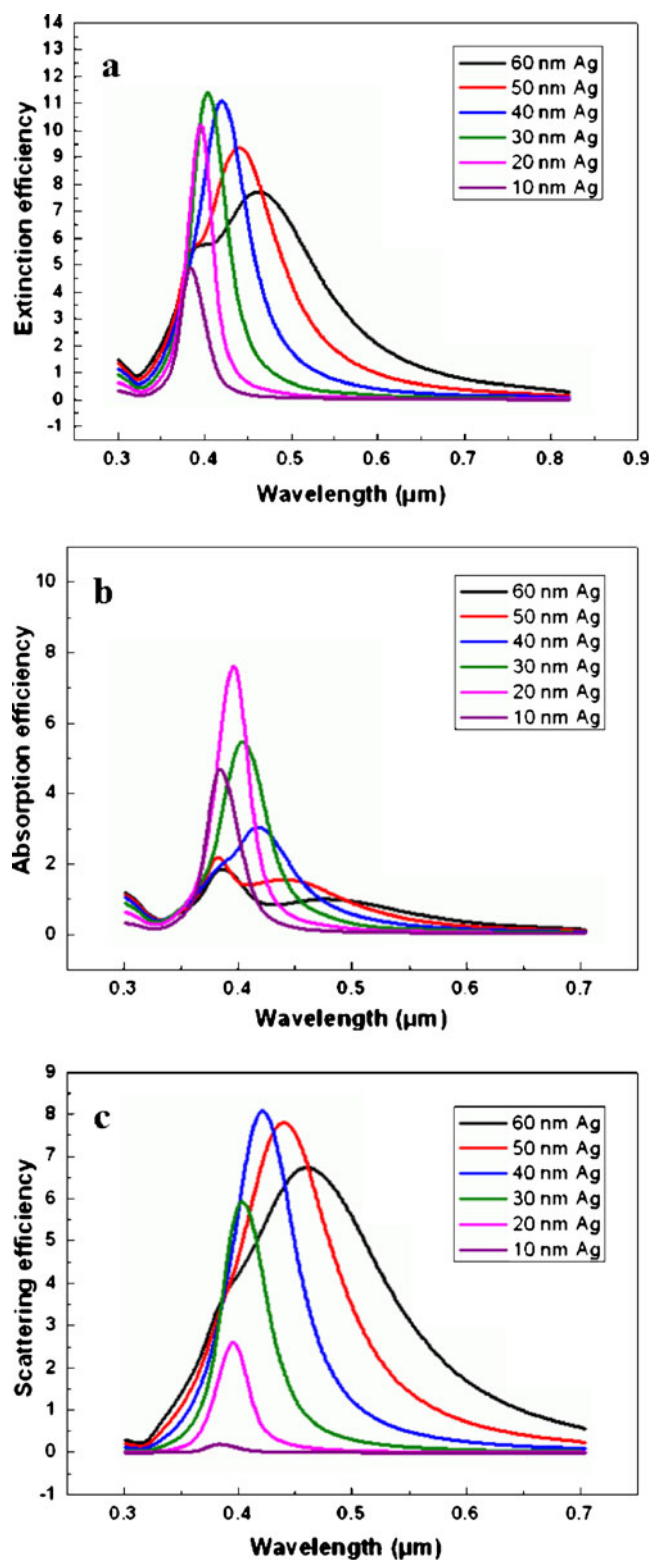


Fig. 2 DDA simulation of optical properties of spherical silver nanoparticles surrounded with the medium, and the refractive index of the medium is 1.33. **a** Extinction efficiencies. **b** Absorption efficiencies. **c** Scattering efficiencies

In Fig. 1, it shows the optical efficiencies of spherical gold nanoparticles surrounded with the medium, and the size of spherical gold nanoparticle is in the range of 10–60 nm. Figure 1a shows the extinction efficiencies of gold nanoparticles. The calculated result in Fig. 1a shows that the extinction efficiencies are strengthening with the increasing sizes of nanoparticles, when the incident wavelength is varied from 300 to 850 nm.

Figure 1b shows the absorption efficiencies of gold nanoparticles. As shown in Fig. 1b, the maximum absorption efficiency of gold nanoparticles is 3.6, when the size of nanoparticle is 50 nm and the incident wavelength is 540 nm. The calculated result also shows that the absorption efficiencies of nanoparticles are strengthening when the size of nanoparticle increases from 10 to 50 nm, and then start to weaken when the size is more than 50 nm. The size of gold nanoparticles can change their absorption efficiencies, and the absorption has been efficient converted to heat energy. It has been reported that the size of gold nanoparticles can affect the heating efficiency [3, 7]. For absorption heating, 50-nm gold nanoparticle is preferred when the incident wavelength is 540 nm because it has the highest absorption efficiency.

Figure 1c shows the scattering efficiencies of gold nanoparticles. The calculated result in Fig. 1c shows that the scattering efficiencies are strengthening with the increasing sizes of nanoparticles as well. For scattering imaging, larger gold nanoparticles are preferred for higher scattering efficiencies [3].

Figure 2 shows the optical efficiencies of spherical silver nanoparticles surrounded with the same medium, and the size of spherical silver nanoparticle is in the range of 10–60 nm as well. Figure 2a shows the extinction efficiencies of silver nanoparticles. The calculated result in Fig. 2a shows that the extinction efficiencies of nanoparticles are strengthening when the size of nanoparticle increases from 10 to 30 nm, and then start to weaken when the size is more than 30 nm, which are different from the extinction spectra of gold nanoparticles as shown in Fig. 1a.

Figure 2b shows the absorption efficiencies of silver nanoparticles. As shown in Fig. 2b, the maximum absorption efficiency of silver nanoparticles is 7.8, when the size of nanoparticle is 20 nm and incident wavelength is 396 nm, which is higher than the maximum absorption efficiency of gold nanoparticles. Figure 2b also shows that the absorption efficiencies of nanoparticles are strengthening when the size of nanoparticle is from 10 to 20 nm, and then begin to weaken when the size is larger than 20 nm. It can be found that 20-nm silver nanoparticle irradiated with 396-nm light has higher absorption heating efficiency than 50-nm gold nanoparticle irradiated with 540-nm light surrounded with the medium, and the refractive index of the medium is around 1.33.

Figure 2c shows the scattering efficiencies of silver nanoparticles. As shown in Fig. 2c, the scattering efficiencies of nanoparticles are strengthening when the size of nanoparticle is in the range of 10–40 nm and then begin to weaken when the size is more than 40 nm, which is different from the strengthening scattering efficiencies of gold nanoparticles with the increasing nanoparticle size. Figure 2c also shows that the maximum scattering efficiency of silver nanoparticles is 8.0, when the size of nanoparticle is 40 nm and the incident wavelength is 420 nm, which is superior to the other sizes of silver nanoparticles and gold nanoparticles in the scattering imaging aspect.

Conclusion

This study displays that gold and silver nanoparticles possess strong absorption efficiencies and scattering efficiencies of visible and near-infrared light. The gold nanoparticle is found to possess the maximum absorption efficiency 3.6 when the size is 50 nm and the incident wavelength is 540 nm and the strengthening scattering efficiency with the increasing size in the medium, and the refractive index of the medium is 1.33. And the silver nanoparticle owns the maximum absorption efficiency 7.8 when the size is 20 nm and the incident wavelength is 396 nm, and the maximum scattering efficiency 8.0 when the size is 40 nm and the incident wavelength is 420 nm in the same medium. The conditions for achieving the maximum absorption efficiency and scattering efficiency can be used for the absorption heating and scattering imaging agents in the future.

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